General Trauma Management

Primary Survey

General Impression
Appearance, Breathing, Activity

Manage life threats as they are identified

Trauma: C - A – B - C
Catastrophic Hemorrhage/C-Spine, Airway, Breathing, Circulation

Catastrophic Hemorrhage

- **Control life-threatening external bleeding** with tourniquets, pressure or hemostatic dressings.
- Manual c-spine motion restriction (if indicated)

Airway

Secure the airway with jaw thrust if cervical spine injury is suspected.

Breathing

Administer Oxygen
- Titrate oxygen to maintain Sp02 >94%.

Ventilation Support
- Support inadequate ventilations with BVM ventilations.

Tension Pneumothorax:
- Perform Chest Needle-Decompression.

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Sucking Chest Wounds
- Apply Chest Seal

Circulation

Unstable Pelvic Fracture
- Apply Pelvic binder

Establish IV Access
- Consider IO Access

Consider ALS Back-Up

- If Advanced Life Support Skills are required
- Consider requirement for narcotic analgesia

Establish Triage Category

Life Threatening
Red Priority

Serious
Yellow Priority

Minor
Green Priority

Rapid Trauma Survey

- Perform a physical examination of the patient to identify major injuries.

General Trauma Management

Adult Fluid Resuscitation

If SBP:
- <90 mmHg, or/
- HR >120 BPM

→ **500mL Fluid Bolus**
- Of Normal Saline or Lactated Ringers IV/IO.
- Repeat fluid bolus' to maintain a SBP of 90mmHg (max 2000mL)

If SBP:
- >90 mmHg, or/
- HR < 120 BPM

→ **No IV Fluids are Required** (monitor closely)

Pediatric Fluid Resuscitation

Tachycardic for age **AND** Signs of poor perfusion

→ **20mL/kg Fluid Bolus**
- Of Normal Saline or Lactated Ringers IV/IO.
- Repeat fluid bolus' to maintain a normal SBP for age (max 60mL/kg)

Consider Tranexamic Acid (TXA)

Adult	**1000mg** IV/IO Dilute in 100ml normal saline given over 10min
Pediatric	**15mg/kg** IV/IO Maximum 1000mg Infuse over 10min

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Avoid Hypothermia
- Remove wet clothing.
- Cover the patient with a thermal (foil) blanket.
- Attempt to limit the time the patient is uncovered for assessment/treatment.

Secondary Survey
- Systematic, detailed head-to-toe physical examination to identify non-life-threatening injuries.
- OPQRST and SAMPLE history

Spinal Motion Restriction
- Assess for Spinal Motion Restriction Criteria

Pain Management
- Assess and manage pain

Minor Injury Management
- Splint fractures or dislocations
- Cover open wounds with sterile dressings.

Patient Reassessment

Frequently Reassess the patient:
- Red Priority: Every 5 minutes.
- Yellow Priority: Every 10 minutes.
- Green Priority: Every 15 minutes.

Patient Care Goals

1. Rapidly identify and manage life-threatening injuries in trauma patients to prevent mortality and minimize morbidity.
2. Stabilize patients through targeted interventions, including hemorrhage control, airway management, and shock prevention, while prioritizing rapid transport to a trauma center.
3. Ensure safe and systematic care delivery, integrating spinal motion restriction, pain management, and targeted care delivery to optimize outcomes.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with a traumatic mechanism of injury.
- Patients with traumatic injuries from blunt, penetrating, or blast mechanisms, including:
    - External or internal hemorrhage (e.g., lacerations, gunshot wounds, or blunt abdominal trauma).
    - Airway or respiratory compromise (e.g., facial fractures, chest trauma).
    - Neurological deficits or altered mental status (e.g., traumatic brain injury, spinal cord injury).
    - Fractures, dislocations, or soft tissue injuries causing pain or instability.

Exclusion Criteria

- Patients with no evidence of trauma

Key Patient Assessment Points

1. **Mechanism of Injury**
    - Identify the type and severity of trauma (e.g., blunt, penetrating, or blast) to predict injury patterns.
    - Assess kinetic energy involved (e.g., speed of vehicle, height of fall, or weapon type) to prioritize occult injuries.
2. **Primary Survey**
    - **Life-Threatening Injury Identification**
        - Detect immediate threats **using the C-ABC algorithm** (Catastrophic Hemorrhage, Airway, Breathing, Circulation).
        - Focus on catastrophic bleeding, airway obstruction, tension pneumothorax, or signs of shock (e.g., pallor, delayed capillary refill).
3. **Neurological and Spinal Assessment**
    - Evaluate for traumatic brain injury (TBI) using Glasgow Coma Scale (GCS) or AVPU, noting pupil asymmetry or posturing.
    - Assess for spinal injury risk (e.g., midline tenderness, neurological deficits) to guide spinal motion restriction (SMR).
4. **Rapid Trauma Survey**
    - Perform a rapid head-to-toe assessment to identify additional injuries (e.g., fractures, open wounds, or deformities).
    - Check for signs of internal injuries (e.g., abdominal rigidity, chest crepitus, or pelvic instability).
5. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Consider ALS Back-Up**
    - Request for ALS back-up as per Universal Care CPG.
        - Consider the requirement for narcotic analgesia.
2. **Primary Survey**
    - Ensure scene safety, don personal protective equipment (PPE), and wear high-visibility apparel if needed.
    - **Apply the C-ABC algorithm:**
        - **Catastrophic Hemorrhage / C-Spine Control:**
            - Control life-threatening external bleeding with tourniquets, pressure or hemostatic dressings.
            - Maintain manual stabilization of c-spine if mechanism of injury suggests c-spine injury.

- **Airway:**
  - If airway obstruction or altered mental status results in the inability to maintain airway:
    - Secure the airway with jaw thrust if cervical spine injury is suspected.
    - See the Airway Management CPG.
- **Breathing:**
  - Consider **Tension Pneumothorax**
    - If tension pneumothorax present, perform Chest Needle-Decompression.
      - Indications:
        - Chest trauma with tracheal deviation
        - Decreased breath sounds on affected side
        - Hypotension
  - If Open/**Sucking Chest Wounds**
    - Apply Chest Seal to open chest wounds.
  - **Administer Oxygen**
    - Titrate oxygen to maintain Sp02 >94%.
  - **Ventilation Support**
    - Support inadequate ventilations with BVM ventilations.
- **Circulation:**
  - **Consider Pelvic Fracture**
    - If pelvis is unstable, apply pelvic binder.
  - **Establish large bore IV/IO access.**
    - Consider IO Access if IV access is delayed or not feasible.
  - **Consider IV/IO fluid administration** (see below).

3. **Rapid Trauma Survey**
   - Conduct a systematic detailed head-to-toe examination physical to identify non-life-threatening injuries (e.g., fractures, lacerations).
4. **Fluid Resuscitation**
   - Note there are additional fluid resuscitation targets for Traumatic Brain Injuries (below)
   - **Adults:**
     - If SBP is less than 90 mmHg or HR greater than 120 BPM:
       - Consider 500 mL bolus of Normal Saline or Lactated Ringers IV/IO.
     - If SBP greater than 90 mmHg and heart rate less than 120 BPM, no IV fluids are required.
     - Repeat 500mL IV fluid bolus' as needed to maintain a SPB of 90mmHg
     - Maximum total IV/IO fluid bolus 2000mL.
   - **Pediatrics:**
     - If tachycardic for age, AND signs of poor perfusion (hypotension for age, greater than 2-second capillary refill, altered mental status, hypoxia, weak pulses, pallor, or mottled/cool skin):
       - Administer 20 mL/kg IV/IO crystalloid bolus and reassess.
       - Repeat 20 mL/kg IV/IO fluid bolus' as needed if signs of shock remain.
       - Maximum total IV/IO fluid bolus of 60mL/kg.
     - Fluid administration target is normal blood pressure for patient age.
5. **Traumatic Brain Injury Fluid Administration Target**
   - If Traumatic Brain Injury is identified:
     - Administer IV/IO fluid bolus as above to target:
       - Adults: SBP of 120mmHg.
       - Pediatrics: normal SBP for age.
6. **Consider Tranexamic Acid (TXA)**
   - Indicated for:
     - Uncontrolled Bleeding, within 3 hours from time of injury.
   - Administer:
     - Adults: 1000mg IV/IO (diluted in 100ml normal saline and administered over 10 minutes)

    - Pediatric: 15mg/kg IO/IO diluted with normal saline to make a total of 20ml and infused over 10 minutes.
  - TXA should be administered as early as possible, ideally within 1 hour from time of injury.
  - Doses should be reduced if the patient is known to have renal impairment.
7. **Avoid Hypothermia**
  - Remove wet clothing.
  - Cover the patient with a thermal (foil) blanket.
  - Attempt to limit the time the patient is uncovered for assessment/treatment.
8. **Secondary Survey**
  - Conduct a systematic, detailed head-to-toe physical examination to identify non-life-threatening injuries (e.g., fractures, lacerations).
  - Obtain OPQRST and SAMPLE history, focusing on mechanism, symptoms, and pre-existing conditions.
9. **Spinal Motion Restriction (SMR)**
  - If patient meets criteria for SMR, follow Spinal Motion Restriction CPG.
10. **Pain Management**
  - Assess for patient pain as per Pain Management CPG.
  - Administer analgesia as per Pain Management CPG.
11. **Fracture and Soft Tissue Management**
  - Splint fractures or dislocations to reduce pain and prevent further injury.
  - Cover open wounds with sterile dressings to minimize infection risk.
12. **Patient Re-Assessment**
  - Frequently Reassess the patient as per Universal Care CPG.
  - Reassess all vital signs and interventions (e.g. vital signs, AVPU, GCS, IV patency).
    - Red Priority: Every 5 minutes.
    - Yellow Priority: Every 10 minutes.
    - Green Priority: Every 15 minutes.
13. **Consider Red Flags**
  - Uncontrolled hemorrhage or signs of hypovolemic shock (e.g., SBP <90 mmHg, tachycardia, delayed capillary refill >2 seconds).
  - GCS ≤8, unequal pupils, or posturing indicating severe TBI.
  - Respiratory distress or hypoxia (SpO2 <90%) suggesting chest trauma (e.g., pneumothorax, hemothorax).
  - Suspected spinal cord injury with neurological deficits (e.g., paralysis, sensory loss).
  - High-risk mechanisms (e.g., ejection from vehicle, blast injury) with occult injuries requiring trauma center evaluation.

Patient Safety Considerations

- **Rapid Transfer Requirements**
  - Patients with major hemorrhage, hemodynamic instability, penetrating torso trauma, require rapid surgical intervention.
  - Minimize scene time scene time as much as possible (goal is under 10 minutes).
- **Elderly Patients** (>65 years):
  - Increased risk of severe injury from low-energy mechanisms (e.g., ground-level falls) due to frailty or anticoagulation.
  - Use cautious fluid resuscitation and monitor for medication interactions (e.g., warfarin, aspirin).
- **Pediatric Patients** (<13 years):
  - Use Broselow Tape for weight estimation and medication dosing to ensure accuracy.
  - Involve a parent or guardian during assessment to reduce anxiety and provide history.
  - Adjust SMR devices and splints for pediatric anatomy to prevent complications.
- **Transport Safety**
  - Prioritize rapid transport for critical patients (red priority) while continuing interventions en route.
  - The safety of the EMS team while driving is paramount, safe driving is required to prevent injury.

Clinical Insights & Additional Guidance

- **Transport Considerations**
  - Transport to a level 1 trauma center for critical patients (e.g., polytrauma, TBI, or hemorrhagic shock).
  - Stable patients with isolated injuries (e.g., single fracture) may be transported to a general hospital if no trauma center is required.
  - Consider Airwing for time-critical cases with ground transport >30 minutes.

Key Documentation Points

- Identification: Patient's name, age, gender.
- Mechanism / Medical Complaint: Description of mechanism of injury.
  - Falls: Estimated height
  - Traffic Accident: approximate speed, and any complications: roll-over, ejection, trapped (time), other deceased or Red Priority in same vehicle.
- Injuries / Information: Describe all injuries found.
- Signs: Primary survey abnormalities & vital signs.
- Treatment: Interventions performed
  - Catastrophic Hemorrhage Control: Tourniquet, hemostatic agents used.
  - C-spine: Spinal Motion Restrictions Used.
  - Airway: Airway adjuncts (OPA/NPA) or interventions
  - Breathing: Oxygen/BVM support
  - Circulation: IV/IO access gained, amount of fluid administered.
  - Medications Administered.
- Allergies: Any known medication allergies.
- Medications: Recent medications or known medical history.
- Background: Relevant medical history.
- Other: Time incident happened.

Key Performance Indicators

1. Apply a tourniquet for patients with uncontrolled limb bleeding.
2. Apply a pelvic binder for patients with an unstable pelvis.
3. Apply a chest seal for patients with open or sucking chest wounds.
4. Request ALS back-up for patients with a GCS<8 of Systolic BP <90 and transfer time of more than 10 minutes.
5. On scene time of <10 minutes for red priority patients.

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